Table 9: Independent Recalculation and Numerical Audit

Revision of the supplied DebunkMeg DURF technical report

30 October 2026

Outcome. The original regional calculation is not reproducible: the six original masks, standardised-pressure stress export and original extraction command are missing. At the user's request, new code now performs a fresh 1 MPa linear-elastic solve and calculates explicitly different geometric-region statistics. The replacement table is not presented as reproduction of the original constitutive study or anatomical annotations.

1. Audited replacement table and conclusion

Table 9: Recalculated region-wise percentile stress and stress-amplification factors under a standardised crown-face pressure $\sigma_0 = 1$ MPa. Regions are newly defined geometric proxies. Values are equal-weight element summaries of σ_{vM} (MPa), not pointwise fracture criteria.

Region	n_e	$Q_{0.90}$	$Q_{0.95}$	$Q_{0.99}$	$\mathrm{SAF}_{0.95}$
Central crown reference R_0	1,186	9.725	11.052	13.140	11.052
Edge-side A R_A	1,606	20.183	22.373	30.605	22.373
Edge-side B R_B	1,050	13.911	16.809	19.610	16.809
Macroscopic apical tip R_{tip}	723	0.409	0.540	0.779	0.540
Transition proxy R_{cr}	21,752	31.806	39.214	56.280	39.214
Root-side band R_{root}	6,801	134.899	188.756	237.625	188.756

n_e is the number of tetrahedra in the mask. For this verified 1 MPa load, SAF is numerically equal to Q in MPa. A/B anatomical mesial/distal identity and the labial/lingual identity of the loaded face remain unverified. The apical mask is the macroscopic tooth tip, not individual serration-tip process zones.

The edge-side-to-central-crown $Q_{0.95}$ ratios are 2.024 and 1.521. In this case the root-side band has the highest regional percentile and the macroscopic apex the lowest. Thus the former "tip process zone highest, root interior lowest" ordering is not supported by this new experiment. The root result is strongly tied to the imposed support/bending configuration. A/B have not been anatomically identified as mesial/distal, and the macroscopic apex mask is not an individual serration-tip process zone.

2. Numerical contract and geometry

The weak form is $\int_\Omega \varepsilon(v) : C : \varepsilon(u)\, dV = \int_{\Gamma_p} v \cdot (-p\mathbf{n})\, dA$, with $u = 0$ on the two root-cap supports. Strains are small, material is homogeneous, and $\lambda = 29$ GPa, $\mu = 28.5$ GPa (the supplied report's Table 4 elastic parameters). These imply $E = 71.374$ GPa and $\nu = 0.252174$. This is a linear-elastic baseline, not a new nonlinear MR5 calibration. The archived refined TetWild mesh uses millimetres according to local reconstruction metadata. All coordinates are multiplied by 10^{-3} before assembling in SI. Its 55,859 input tetrahedra include six face-connected components. Seven off-main-component tetrahedra, volume fraction 2.996738×10^{-6}, were excluded; 13,584 nodes and 55,852 tetrahedra remain. No stress-based geometry selection is performed. Original element IDs and the removal map are archived. Exact integrated solid-volume second moments define the PCA frame, avoiding dense-vertex sampling bias. Longitudinal $s = 0$ is the macroscopic apex and $s = 1$ the root end. The sign follows original scan $+Z$. The positive thickness direction follows scan $+X$, without independently establishing labial/lingual identity. The local silhouette coordinate r is derived from 40 longitudinal geometry bins; ζ is normalised thickness. Masks are generated and saved before solving.

3. Exact region, pressure and support rules

Region	Rule
R_0	$0.25 \le s \le 0.55$, $\lvert r\rvert \le 0.35$, $\lvert\zeta\rvert \le 0.5$
R_A	$0.12 < s < 0.64$, $r \le -0.78$
R_B	$0.12 < s < 0.64$, $r \ge 0.78$
R_{tip}	$s \le 0.12$ (macroscopic apex)
R_{cr}	$0.64 \le s \le 0.76$ (transition proxy)
R_{root}	$0.78 \le s \le 0.84$ (below the root supports)

The 1 MPa pressure patch contains exterior faces whose centroids satisfy $0.18 \le s \le 0.64$, $|r| \le 0.75$, with outward-normal alignment at least 0.5 to the positive thickness direction. Each face contributes $-pA\mathbf{n}/3$ to each of its three nodes. The 2,849 loaded triangles have total area 418.091454 mm^2 and net force $(-392.694929, -95.707621, -12.678506)$ N in scan coordinates. Each root lobe is capped relative to its own maximum s, separated by transverse sign. Fixed nodes satisfy $s \ge 0.80$ and $s \ge s_{\max}(\text{side}) - d$, with baseline $d = 0.04$. All three translations are fixed. Both lobes must contain fixed nodes, and no fully clamped tetrahedron may enter R_{root}. The baseline has 169 fixed nodes and 40,245 free displacement DOFs. This idealised attachment is not a validated periodontal model.

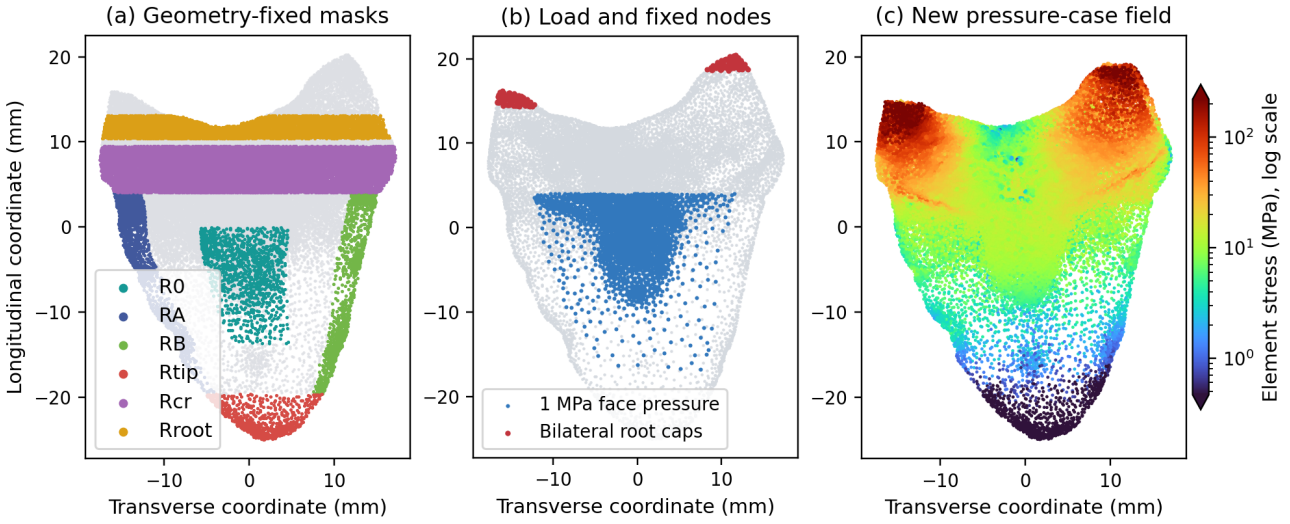


Figure A1. Geometry-only masks, bilateral root supports, pressure patch and new element stress field. Stress colours use a logarithmic scale clipped to the global 2nd–99th percentiles for display. All numerical tables use the full un-clipped field.

4. Numerical implementation and validation

Four-node tetrahedra use constant strain $B_e u_e$ and $K_e = V_e B_e^T C B_e$. Surface normals are oriented away from each owning tetrahedron. The sparse Dirichlet system is solved with SciPy SuperLU, with residual-based iterative refinement if needed. Stress is evaluated directly per element, without nodal smoothing or nearest-neighbour transfer. All element stresses are finite and nonnegative.

Check	Observed error (relative unless stated)
Five-tetrahedron cube: affine displacement / constant-stress traction patch	3.963×10^{-15}
Rigid-body strain (maximum absolute strain)	3.469×10^{-18}
Stiffness symmetry	8.299×10^{-17}
Main solve: free-DOF residual	5.109×10^{-11}
Main solve: total force balance	1.425×10^{-12}
Main solve: total torque balance	5.790×10^{-13}
Main solve: $u^T K u = u^T f$ energy identity	2.876×10^{-12}
Second solve with doubled pressure: displacement scaling	0

Main maximum displacement is 0.436472 mm and strain energy is 0.0435872 J. The tests check finite-system implementation and equilibrium; they do not validate the biological material, load, anatomy, fracture mechanism or mesh convergence. The independent quantile check sorts each sample using Python and evaluates the interpolation formula separately from NumPy.

5. Quantile convention and bounded sensitivity checks

For sorted $v_0 \le \cdots \le v_{n-1}$, let $h = (n-1)q$, $j = \lfloor h \rfloor$ and $\alpha = h - j$. The primary table uses $Q_q = (1-\alpha)v_j + \alpha v_{\lceil h \rceil}$ with equal element weights. This agrees with NumPy's explicit `method='linear'` convention. SAF divides this stress by the verified 1 MPa applied pressure. The crown reference stress is not the SAF denominator. The volume-weighted comparison uses the inverse cumulative element-volume CDF; it deliberately answers a different sampling-measure question. Equal-element and volume-weighted results differ because this mesh has locally variable element density. Neither is automatically mesh convergent.

Region suffix	Equal-element $Q_{0.95}$ MPa	Volume-weighted $Q_{0.95}$ MPa	Root-cap sensitivity range MPa
0	11.052	8.281	11.047–11.214
A	22.373	20.788	22.373–22.888
B	16.809	14.066	16.116–16.809
tip	0.540	0.665	0.529–0.543
cr	39.214	36.191	39.161–39.523
root	188.756	147.011	176.883–190.933

The last column varies bilateral cap depth $d = 0.03, 0.04, 0.05$ on the same mesh. Edge-mask threshold sensitivity uses $|r| = 0.70, 0.78, 0.86$: side A spans 21.756–23.503 MPa; side B spans 14.237–18.126 MPa. These checks do not constitute physical-mask validation or mesh refinement.

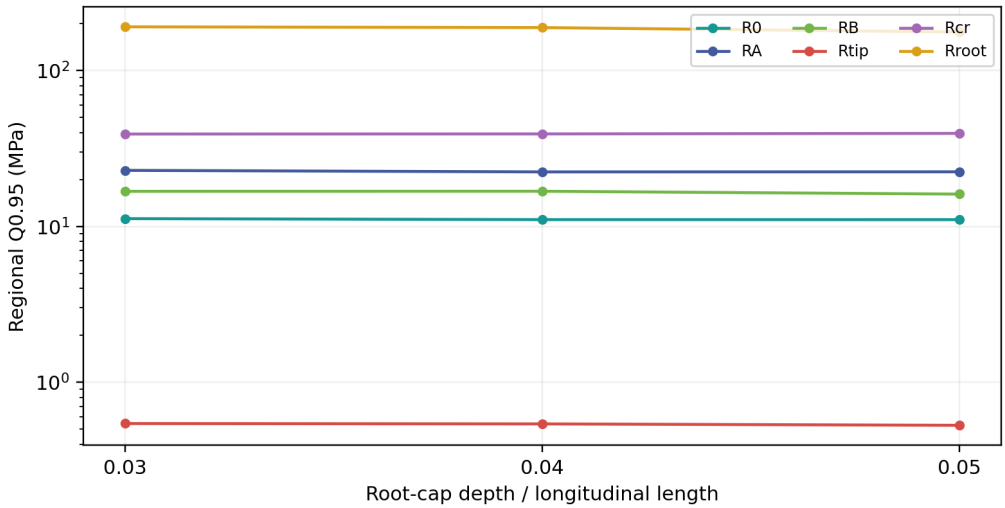


Figure A2. Regional $Q_{0.95}$ for three bilateral cap depths. The regional ordering is unchanged in these three cases; that observation is confined to this small tested boundary-condition family.

6. Audit of archived data and former Table 9

The old smoothed, uncompressed and raw-tet stress files are COMSOL exports of `solid.mises` in Pa. Direct re-extraction gives the following *whole-export sample* statistics; these are not regional values or the fresh pressure solve.

Export	Samples	$Q_{0.90}$ MPa	$Q_{0.95}$ MPa	$Q_{0.99}$ MPa	Maximum MPa
rawtet	24,376	0.223429	0.250069	0.289708	0.384152
smoothed	191,031	0.224088	0.253390	0.296859	0.401620
uncompressed	191,202	0.224401	0.253628	0.298302	0.406188

All three exports contain only finite, nonnegative stress values; data row counts agree with their headers. Seventeen of the 18 former regional Q values exceed each archived file's global maximum. They cannot be extracted from those same unscaled samples. This does not disprove a different unarchived experiment; it establishes that the available archive cannot support the old table. The available export program `AddVonMisesPointCloudToStaticDynamics.java` configures MR5 parameters, disables fixed supports and boundary pressure features, adds rigid-motion suppression and a 5×10^4 N/m^3 body load, runs `std_mr5`, and exports `solid.mises` from `dset4`. That source is not a 1 MPa crown-face pressure protocol. The text headers name `static_dynamics.mph`; the old frontal plot instead labels a later `strict3bdf.mph` model. The archive does not prove an immutable study/export linkage, so these source observations are not elevated to certification of every historic COMSOL setting. **Arithmetic audit.** All six old $\mathrm{SAF}_{0.95}$ entries equal their listed $Q_{0.95}$ in MPa, which is the correct arithmetic *if* the pressure really is 1 MPa. This does not validate either the field or the masks. Relative to the old central-crown 0.89 MPa value, the old mesial/distal/tip ratios would be 24.494, 23.483 and 51.910, respectively; SAF and crown-reference ratio must not be conflated. The old expression $0.8/[52\sqrt{\pi(0.5 \times 10^{-3})}]$ gives approximately 0.388 MPa, so the rounding to 0.39 MPa is arithmetically valid. Its assumed SAF was not validated, and von Mises stress is not itself a calibrated mode-I crack-driving stress. No biological critical pressure is established or inserted into the revised table.

7. Provenance, reproduction and scope

`run/manifest.json` is a validated version-2 record with status `completed`, input hashes before/after, software versions, execution command, checks and output hashes. Its validation concerns provenance, not scientific universality. The actual calculation command, from the package root, is:

```
python3 scripts/recalculate.py --output reproduced_run
```

The saved masks refer to zero-based tetrahedron IDs in the archived solution. A separate index array maps retained elements back to the original input ordering. The JSON results record mesh and mask hashes; each old export is hashed separately. Exact file names and commands are documented in the accompanying README. No random sampling is used. This remains one mesh, one homogeneous linear material and a small boundary/mask sensitivity set. The original anatomical masks, original standardised-pressure study, mesial/distal assignment, individual serration apex regions, validated labial orientation, physical root attachment and biological fracture data have not been recovered. New anatomy-specific annotations and at least three compatible mesh levels are needed before stronger claims. The revised 26-page manuscript preserves unchanged content as PDF backgrounds. All changed wording and Table 9 are editable LaTeX. It is *not* a recovered fully editable original manuscript source.